



AI Machine Learning Analyst

QP Code: SSC/Q8113

Version: 4.0

NSQF Level: 5

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Qualification Pack

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Qualification Pack

SSC/Q8113: AI Machine Learning Analyst

Brief Job Description

A Machine Learning Analyst is crucial in designing, developing, deploying, and maintaining machine learning models that transform data into actionable insights for addressing business challenges. This professional combines expertise in software engineering, data analysis, and machine learning principles to create systems capable of learning from data and making predictions. Key responsibilities include preparing and processing data to ensure high quality, selecting and creating features, exploring various algorithms for specific tasks, and implementing neural networks using appropriate frameworks. Model training and evaluation are crucial, utilizing hyperparameter tuning and metrics such as accuracy and F1 score to ensure performance. After deployment, the engineer establishes monitoring systems to track performance, data quality, and utilization, identifying signals for model retraining and implementing automated pipelines.

Personal Attributes

A Machine Learning engineer needs to have strong analytical and logical skills, attention to detail and problem solving ability. S/he needs to have good communication skills to work with various stakeholders that includes Data Scientists, Data Architects, Business Analysts etc. to ensure alignment on business objectives

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8161: Handle and preprocess data](#)
2. [SSC/N8121: Develop & deploy the machine learning models](#)
3. [SSC/N8122: Monitor and maintain machine learning models post-deployment](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
Country	India

Qualification Pack

NSQF Level	5
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2512.NIL
Minimum Educational Qualification & Experience	Completed 3-year diploma (after 10th) (PG/UG/diploma with courses related to Engg./ Science) with 1.5 years of experience Relevant Experience- AI/Computer Science/IT. The relevant experience would include work, internship and apprenticeship undertaken post-completion of relevant educational qualification OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience Relevant Experience- AI/Computer Science/IT. The relevant experience would include work, internship and apprenticeship undertaken post-completion of relevant educational qualification OR Completed 2nd year of UG (UG Diploma) (PG/UG/diploma with courses related to Engg./ Science)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	4.0
Reference code on NQR	QG-05-IT-045832025-V2-NASSCOM
NQR Version	2

Qualification Pack

SSC/N8161: Handle and preprocess data

Description

This Nos is about data handling and preprocessing

Scope

The scope covers the following :

- Problem definition, Data collection and handling, Data exploration, Data cleaning, Data Sampling, Data pipeline development, Version control of datasets, Documentation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Collaborate with stakeholders to define the business problem and objectives.
- PC2.** Understand the importance of data quality and integrity
- PC3.** Demonstrate methods to collect data from various sources
- PC4.** Gather data from various sources relevant to the business problem and merge it into single coherent dataset
- PC5.** Conduct exploratory data analysis (EDA) to understand the data structure and statistical properties.
- PC6.** . Identify and rectify errors in the data such as missing values and duplicates
- PC7.** Select a representative subset of data for analysis or training models
- PC8.** Understand how to develop efficient and reproducible data pipelines.
- PC9.** Maintain and track the evolution of datasets
- PC10.** Document the data handling processes and decisions made during preprocessing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Follow organizational policies and procedures for defining documenting and sharing data governance processes involving relevant stakeholders and using standard templates and tools.
- KU2.** Learn how to identify and categorize internal and external data assets based on business requirements and apply performance measures to maintain data quality and integrity.
- KU3.** Develop and implement standards for data utilization and governance structures to ensure consistent data management and compliance.
- KU4.** Understand how to work across various operating systems such as Linux Ubuntu and Windows for effective data management and governance tasks.
- KU5.** Recognize the importance of involving the right stakeholders when establishing data governance processes and documenting standards to support organizational data strategy.

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make decisions on suitable course of action
- GS2.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Collaborate with stakeholders to define the business problem and objectives.	3	5	-	2
PC2. Understand the importance of data quality and integrity	3	5	-	2
PC3. Demonstrate methods to collect data from various sources	3	5	-	2
PC4. Gather data from various sources relevant to the business problem and merge it into single coherent dataset	3	5	-	2
PC5. Conduct exploratory data analysis (EDA) to understand the data structure and statistical properties.	3	5	-	2
PC6. . Identify and rectify errors in the data such as missing values and duplicates	3	5	-	2
PC7. Select a representative subset of data for analysis or training models	3	5	-	2
PC8. Understand how to develop efficient and reproducible data pipelines.	3	5	-	2
PC9. Maintain and track the evolution of datasets	3	5	-	2
PC10. Document the data handling processes and decisions made during preprocessing	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8161
NOS Name	Handle and preprocess data
Sector	IT-ITeS
Sub-Sector	
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8121: Develop & deploy the machine learning models

Description

This NOS is about developing and deploying the machine learning models

Scope

The scope covers the following :

- Data Preparation for Modeling, Feature engineering, Model development, Training and Tuning, Evaluation, testing and refinement, Deployment

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Ensure data is properly prepared by cleaning and transforming raw data to make it suitable for training and validation.
- PC2.** Select or create features that enhance model performance based on the data and the specific problem context.
- PC3.** Explore various machine learning algorithms including supervised unsupervised and reinforcement learning to determine the best fit for the task
- PC4.** Design the model architecture especially for deep learning models by defining layers nodes and activation functions and develop or customize algorithms if needed
- PC5.** Build and train the models using training data and optimize parameters through techniques like grid search or random search to improve accuracy and generalization
- PC6.** Evaluate the effectiveness of the models using metrics such as accuracy precision recall ROC AUC and confusion matrix to assess performance thoroughly
- PC7.** Apply techniques such as K fold cross validation and regularization to ensure the robustness of the models and prevent overfitting
- PC8.** Improve the models iteratively based on evaluation results and consider combining multiple models through methods like bagging or boosting to enhance performance.
- PC9.** Transition the trained models into production by building APIs or integrating them into existing applications to ensure seamless deployment.
- PC10.** Implement continuous integration and continuous deployment practices to automate model deployment and updates and enable other applications to interact with the models for real-time predictions.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the purpose and aims of the system being developed and follow organizational policies for evaluating the technical performance of algorithmic models
- KU2.** Learn how to access and interpret data sources and documents while understanding the organization policies for sharing and documenting algorithmic models

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- KU3.** Identify and analyze different algorithmic models data flows and data structures to ensure they meet system requirements and limitations
- KU4.** Evaluate algorithmic models for speed runtime memory dependencies and performance using methods such as Big O notation and identify anomalies or issues
- KU5.** Use standard templates and tools to document models and performance evaluations work across various operating systems and involve the right stakeholders in performance assessments

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make decisions on suitable course of action
- GS2.** plan and organize your own work to achieve targets and deadlines
- GS3.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Ensure data is properly prepared by cleaning and transforming raw data to make it suitable for training and validation.	3	5	-	2
PC2. Select or create features that enhance model performance based on the data and the specific problem context.	3	5	-	2
PC3. Explore various machine learning algorithms including supervised unsupervised and reinforcement learning to determine the best fit for the task	3	5	-	2
PC4. Design the model architecture especially for deep learning models by defining layers nodes and activation functions and develop or customize algorithms if needed	3	5	-	2
PC5. Build and train the models using training data and optimize parameters through techniques like grid search or random search to improve accuracy and generalization	3	5	-	2
PC6. Evaluate the effectiveness of the models using metrics such as accuracy precision recall ROC AUC and confusion matrix to assess performance thoroughly	3	5	-	2
PC7. Apply techniques such as K fold cross validation and regularization to ensure the robustness of the models and prevent overfitting	3	5	-	2
PC8. Improve the models iteratively based on evaluation results and consider combining multiple models through methods like bagging or boosting to enhance performance.	3	5	-	2
PC9. Transition the trained models into production by building APIs or integrating them into existing applications to ensure seamless deployment.	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Implement continuous integration and continuous deployment practices to automate model deployment and updates and enable other applications to interact with the models for real-time predictions.	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8121
NOS Name	Develop & deploy the machine learning models
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	5
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8122: Monitor and maintain machine learning models post-deployment

Description

This Nos is about Monitoring the models post deployment

Scope

The scope covers the following :

- Setup monitoring systems,Logging Predictions and Events,Detect Data Drift,Monitor Model Performance,Alerting and Reporting,User Feedback collection,Conducting Regular Model Audits,Retraining and Updating Models,Maintenance of Data Pipelines,Model Versioning and Rollback,Performance Fine-Tuning,Risk Management and Compliance

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Implement systems to continuously monitor the performance of deployed machine learning models.
- PC2.** Use monitoring tools to collect and visualize key performance metrics such as accuracy and latency.
- PC3.** Define relevant key performance indicators and set up alerts for issues like performance drops or increased prediction latency
- PC4.** Log prediction inputs outputs and metadata to facilitate troubleshooting and diagnostics.
- PC5.** Monitor for changes in data distribution over time using statistical tests and evaluate their impact on model performance
- PC6.** Regularly assess the model's predictive accuracy and other metrics to ensure it maintains expected performance levels
- PC7.** Create feedback channels to gather qualitative insights from users and compare predictions against actual outcomes for accuracy.
- PC8.** Analyze errors such as false positives and false negatives to identify limitations and areas for improvement
- PC9.** Automate data collection preprocessing and retraining pipelines to keep the model current and prevent performance decay
- PC10.** Track changes using tools like Git and DVC and establish procedures for rolling back to previous versions if necessary

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the purpose and aims of the autonomous system being developed and follow organizational policies and procedures for developing software code.

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- KU2.** Learn how to access and interpret data sources and documents while adhering to organizational policies for sharing and documenting code.
- KU3.** Identify and analyze different data flows and structures and evaluate algorithmic models and their designs for system requirements and limitations.
- KU4.** Use code generation and unit testing tools to develop software code create and review test cases and determine component reusability.
- KU5.** Monitor and resolve problems and defects by recording corrective actions working on various operating systems and involving the right stakeholders in development and review processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** apply problem-solving approaches in different situations
- GS2.** analyze data and activities
- GS3.** configure data and disseminate relevant information to others
- GS4.** apply balanced judgments to different situations
- GS5.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Implement systems to continuously monitor the performance of deployed machine learning models.	3	5	-	2
PC2. Use monitoring tools to collect and visualize key performance metrics such as accuracy and latency.	3	5	-	2
PC3. Define relevant key performance indicators and set up alerts for issues like performance drops or increased prediction latency	3	5	-	2
PC4. Log prediction inputs outputs and metadata to facilitate troubleshooting and diagnostics.	3	5	-	2
PC5. Monitor for changes in data distribution over time using statistical tests and evaluate their impact on model performance	3	5	-	2
PC6. Regularly assess the model's predictive accuracy and other metrics to ensure it maintains expected performance levels	3	5	-	2
PC7. Create feedback channels to gather qualitative insights from users and compare predictions against actual outcomes for accuracy.	3	5	-	2
PC8. Analyze errors such as false positives and false negatives to identify limitations and areas for improvement	3	5	-	2
PC9. Automate data collection preprocessing and retraining pipelines to keep the model current and prevent performance decay	3	5	-	2
PC10. Track changes using tools like Git and DVC and establish procedures for rolling back to previous versions if necessary	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8122
NOS Name	Monitor and maintain machine learning models post-deployment
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence & Big Data Analytics
NSQF Level	5
Credits	4
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8161.Handle and preprocess data	30	50	-	20	100	30
SSC/N8121.Develop & deploy the machine learning models	30	50	-	20	100	30
SSC/N8122.Monitor and maintain machine learning models post-deployment	30	50	-	20	100	30
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	110	180	-	60	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.